Lab Assignment no. 2

PRACTICAL NO. 3

NAME: SOHAM SANJAY LOKHANDE

PRN:202501070167

BRANCH : ENTC

DIVISION :ET22

SUBJECT: EDS LAB

3.1.1.NUMPY ARRAY OPERATIONS

Write a Python program to create a NumPy array based on user input and display the following:

- **The created NumPy array.**

- **The number of dimensions of the array using ndim**

- **The shape of the array using shape**
- **The total number of elements in the array using size**

Assume all input elements are valid integers.

Input Format:

- **The first line contains two space-separated integers, *r* and *c*, representing the number of rows and the number of columns.**
- **The next *r* lines contain *c* space-separated integers representing the elements of the array, entered row by row.**

Output Format:

- **The created NumPy array (printed as a 2D array).**
- **An integer representing the number of dimensions of the array.**
- **A tuple representing the shape of the array.**
- **An integer representing the total number of elements in the array.**

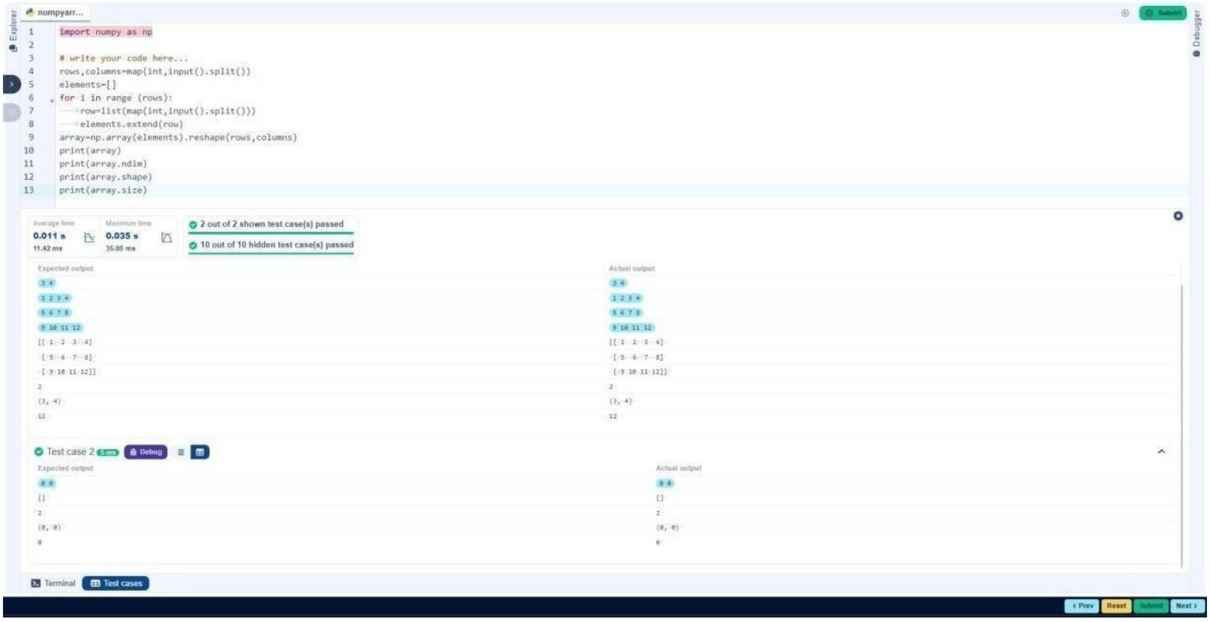
Note: Use reshape() function to reshape the input array with the specified number of rows and columns.

Code:

```
import numpy as np

rows,columns=map(int,input().split()) elements=[]

for i in range (rows):

row=list(map(int,input().split()))

elements.extend(row)

array=np.array(elements).reshape(rows,columns)
```

print(array) print(array.ndim) print(array.shape)

print(array.size)



3.2.1.NUMPY: MATRIX OPERATIONS

The given code takes two 3 x 3 matrices, matrix_a, and matrix_b, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition (matrix_a + matrix_b)**
2. **Subtraction (matrix_a - matrix_b)**
3. **Element-wise Multiplication (matrix_a * matrix_b)**
4. **Matrix Multiplication (matrix_a . matrix_b)**
5. **Transpose of Matrix A**

Input Format:

- **The user will input 3 rows for matrix_a, each containing 3 integers separated by spaces.**

Similarly, the user will input 3 rows for matrix_b, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. **The result of Addition.**
2. **The result of Subtraction.**
3. **The result of Element-wise Multiplication.**
4. **The result of Matrix Multiplication.**
5. **The Transpose of Matrix A.**

Code:

```
import numpy as np


# Input matrices print("Enter Matrix A:") matrix_a = np.array([list(map(int,

input().split())) for i in range(3)]) print("Enter Matrix B:") matrix_b =

np.array([list(map(int, input().split())) for i in range(3)])



# Addition print("Addition (A +

B):")

add=np.array(matrix_a+matrix_b) print(add)

# Subtraction print("Subtraction (A - B):")

sub=np.array(matrix_a-matrix_b) print(sub)

# Multiplication (element-wise) print("Elementwise

Multiplication (A * B):")
```

multi=np.array(matrix_a*matrix_b) print(multi) # Matrix

multiplication (dot product) print("A dot B:")

trans=(matrix_a.dot(matrix_b)) print(trans) #

Transpose print("Transpose of A:")

tran=matrix_a.transpose() print(tran)

3.2.2.NUMPY: HORIZONTAL VS VERTICAL STACKING OF ARRAYS

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking: Stack the two matrices horizontally (side by side).**

- **Vertical Stacking: Stack the two matrices vertically (one below the other).**

Input Format:

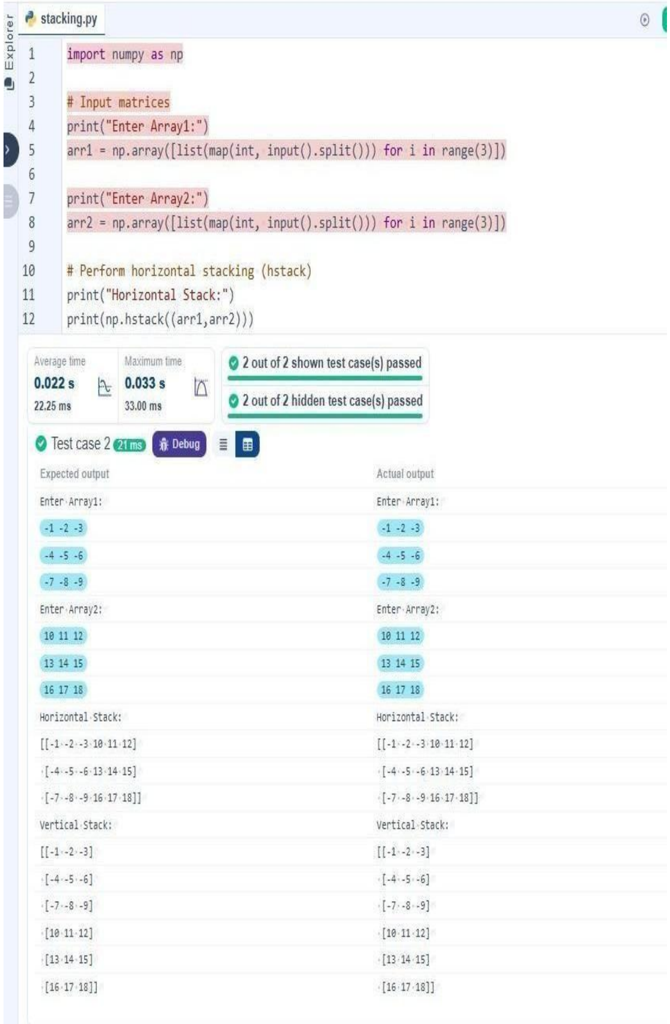
- **The program should first prompt the user to input two 3x3 arrays.**
- **Each array consists of 3 rows, and each row contains 3 space-separated integers.**
- **The user will input the two arrays row by row.**

Output Format:

- **The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.**
- **The program should then display the result of the Vertical Stack (one below the other) of the two arrays.**

Code:

```
import numpy as np

# Input matrices print("Enter Array1:") arr1 = np.array([list(map(int,
input().split())) for i in range(3)])

print("Enter Array2:") arr2 = np.array([list(map(int, input().split()))
for i in range(3)])

# Perform horizontal stacking (hstack) print("Horizontal
Stack:") print(np.hstack((arr1,arr2))) print("Vertical
Stack:")
```



3.2.3.NUMPY:CUSTOM SEQUENCE GENERATION

Write a Python program that takes the following inputs from the user:

- **Start value: The starting point of the sequence.**
- **Stop value: The sequence should end before this value.**
- **Step value: The increment between each number in the sequence.**

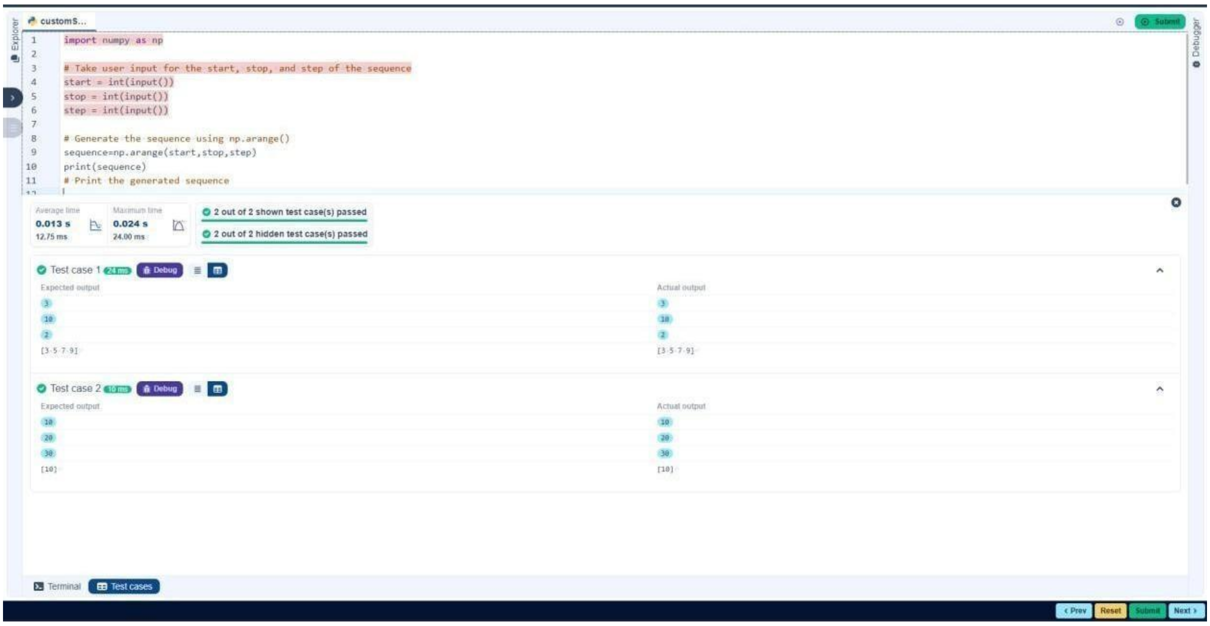
The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- **The user will input three integer values: start, stop, and step, each on a new line.**

Output Format:

The program should print the generated sequence based on the input values.



<u>3.2.4.NUMPY: ARITHMETIC AND STATISTICAL OPERATIONS, MATHEMATICAL OPERATIONS, BITWISE OPERATORS</u>

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- **Compute the element-wise sum, difference, and product of the two arrays.**

2. Statistical Operations:

- **Calculate the mean, median, and standard deviation of array A.**

3. Bitwise Operations:

- **Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).**

Input Format:

- **The first line contains space-separated integers representing the elements of array A.**
- **The second line contains space-separated integers representing the elements of array B.**

Output Format:

- **For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.**

Code:

import numpy as np

def array_operations(A, B):

Convert A and B to NumPy

arrays x=np.array(A) y=np.array(B) #

Arithmetic Operations sum_result

= x+y diff_result = x-y prod_result

= x*y

Statistical Operations

mean_A = np.mean(A) median_A

= np.median(A) std_dev_A =

np.std(A)

Bitwise Operations and_result =

np.bitwise_and(x,y) or_result = np.bitwise_or(x,y)

xor_result

•

= np.bitwise_xor(x,y)

Output results with one space between each element print("Elementwise Sum:", ' '.join(map(str, sum_result))) print("Element-wise Difference:", ' '.join(map(str, diff_result))) print("Element-wise Product:", ' '.join(map(str, prod_result)))

print(f"Mean of A: {mean_A}")   print(f"Median of

A: {median_A}")   print(f"Standard

Deviation of A: {std_dev_A}")

print("Bitwise AND:", ' '.join(map(str, and_result)))   print("Bitwise

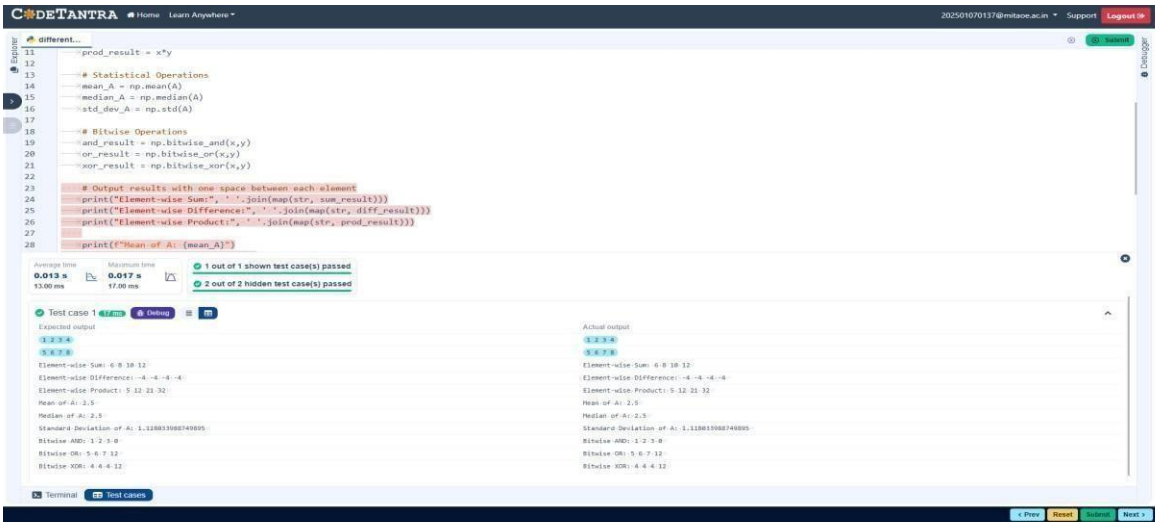
OR:", ' '.join(map(str, or_result)))   print("Bitwise XOR:", '

'.join(map(str, xor_result)))

A = list(map(int, input().split())) # Elements of array A B

= list(map(int, input().split())) # Elements of array B

array_operations(A, B)



3.2.5.NUMPY: COPYING AND VIEWING ARRAYS

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- **Creating a view of the original_array and assigning it to view_array.**
- **Creating a copy of the original_array and assigning it to copy_array.**

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After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- **A single line of space-separated integers.**

Output Format:

- **After modifying the view:**

Original array after modifying view: <original_array>

View array: <view_array>

- **After modifying the copy:**

Original array after modifying copy: <original_array>

Copy array: <copy_array>

Code: import numpy

as np

inputlist = list(map(int,input().split(" ")))

Original array original_array = np.array(inputlist)

Create a view view_array

=original_array.view()

Create a copy copy_array

=original_array.copy()

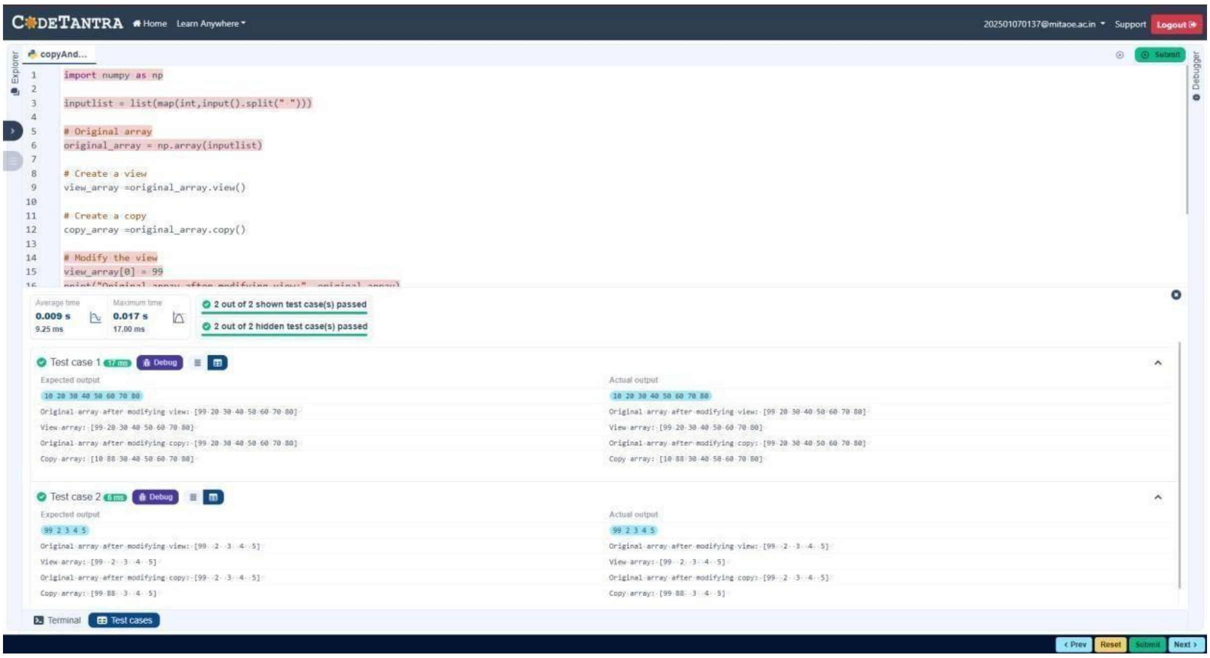
Modify the view view_array[0] = 99 print("Original array after

modifying view:", original_array) print("View array:", view_array)

Modify the copy copy_array[1] = 88 print("Original array after

modifying copy:", original_array) print("Copy array:",

copy_array)



<u>3.2.6.NUMPY: SEARCHING, SORTING, COUNTING, BROADCASTING</u>

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- **search_value: The value to search for in the array.**
- **count_value: The value to count its occurrences in the array.**
- **broadcast_value: The value to add for broadcasting across the array.**

You need to complete the code to perform the following operations:

1. Searching: Find the indices where search_value appears in array1 and print these indices.

2. Counting: Count how many times count_value appears in array1 and print the count.

3. **Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.**
4. **Sorting: Sort array1 in ascending order and print the sorted array.**

Input Format:

1. **A single line containing space-separated integers representing array1.**
2. **An integer search_value represents the value to search for in the array.**
3. **An integer count_value represents the value to count in the array.**
4. **An integer broadcast_value represents the value to add to each element of the array.**

Output Format:

1. **The indices where search_value occurs in array1.**
2. **The count of occurrences of count_value in array1.**
3. **The array after adding the broadcast_value to each element.**
4. **The sorted array.**

Code:

```
import numpy as np


# Input array from the user array1 = np.array(list(map(int,
input().split())))


# Searching search_value = int(input("Value to
search: ")) count_value = int(input("Value to
count: ")) broadcast_value = int(input("Value to
add: "))
```

Find indices where value matches in array1

a=np.where(array1==search_value)[0]

print(a)

Count occurrences in arraycount_nonzero(arrau1==count_value)

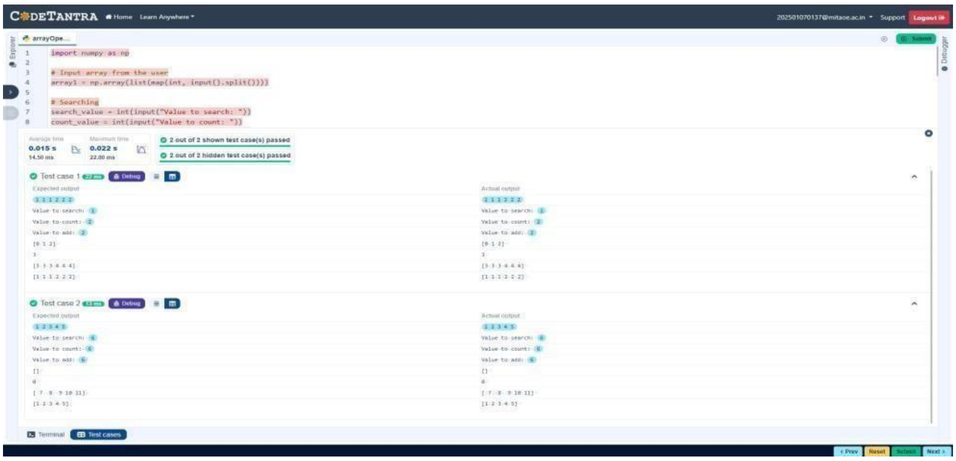
b=np.count_nonzero(array1==count_value)

print(b)
# Broadcasting addition c=array1

+ broadcast_value print(c)

Sort the first array

d=np.sort(array1) print(d)



3.2.7.STUDENT DATA ANALYSIS AND OPERATIONS

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.**
- **Find total students: Determine the total number of students in the dataset.**
- **Print all student roll numbers: Extract and print the roll numbers of all students.**
- **Print Subject 1 marks: Extract and print the marks of all students in Subject 1.**
- **Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.**

- **Find maximum marks in Subject 3: Identify the highest marks in Subject 3.**
- **Print all subject marks: Display the marks of all students for each subject.**
- **Find total marks of students: Compute the total marks for each student across all subjects.**
- **Find the average marks of each student: Compute the average marks for each student.**
- **Find average marks of each subject: Compute the average marks for all students in each subject.**
- **Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.**
- **Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.**
- **Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.**
- **Find the roll number of the student with minimum marks in Subject 2: Identify the student with the lowest marks in Subject 2 and print their roll number.**
- **Find the roll number of students who scored 24 marks in Subject 2: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.**
- **Find the count of students who got less than 40 marks in Subject 1: Count the number of students who scored less than 40 marks in Subject 1.**
- **Find the count of students who got more than 90 marks in Subject 2: Count the number of students who scored more than 90 marks in Subject 2.**
- **Find the count of students who scored >=90 in each subject: Count the number of students who scored 90 or more marks in each subject.**
- **Find the count of subjects in which each student scored >=90: Determine how many subjects each student scored 90 or more marks in.**
- **Print Subject 1 marks in ascending order: Sort and print the marks of students in Subject 1 in ascending order.**
- **Print students who scored between 50 and 90 in Subject 1: Display students who scored marks between 50 and 90 in Subject 1.**
- **Find index positions of students who scored 79 in Subject 1: Identify the index positions of students who scored exactly 79 marks in Subject 1.**

Code:

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# 1. Print all student details print("All student
Details:\n", a  )

# 2. print total students

print("Total Students:",a.shape[0] )

# 3. Print all student Roll numbers print("All
Student Roll Nos",  a[:,0] )

# 4. Print subject 1 marks  print("Subject
1 Marks", a[:,1])

# 5. print minimum marks of Subject 2 print("Min marks
in Subject 2", np.min(a[:,2] ))

# 6. print maximum marks of Subject 3 print("Max marks in
Subject 3", np.max(a[:,3]) )

# 7. Print All subject marks print("All subject
marks:", a[:,1:])
```

```
# 8. print Total marks of students print("Total
Marks", np.sum(a[:,1:],axis=1) )

# 9. print average marks of each student avg=np.mean(a[:,1:],axis=1)
print(np.round(avg,1))

# 10. print average marks of each subject print("Average
Marks of each subject", np.mean(a[:,1:],axis=0) )

# 11. print average marks of S1 and S2 print("Average Marks of
S1 and S2", np.mean(a[:,1:3],axis=0 ) )

# 12. print average marks of S1 and S3 print("Average Marks of
S1 and S3",  np.mean(a[:,[1,3]],axis=0) )

# 13. print Roll number who got maximum marks in Subject 3 i=np.argmax(a[:,3]) print("Roll no
who got maximum marks in Subject 3",  a[i,0]  )

# 14. print Roll number who got minimum marks in Subject 2
j=np.argmin(a[:,2]) print("Roll no who got minimum marks in Subject
2",  a[j,0] )

# 15. print Roll number who got 24 marks in Subject 2

print(f"Roll no who got 24 marks in Subject 2 [{a[a[:,2]==24,0]}]" )

# 16. print count of students who got marks in Subject 1 < 40 print("Count of
students who got marks in Subject 1 < 40",   np.sum(a[:,1]<40)  )
```

17. print count of students who got marks in Subject 2 > 90 print("Count of

students who got marks in Subject 2 > 90:",   np.sum(a[:,2]>90))

Count=[len(np.argwhere(a[:,1]>=90)),len(np.argwhere(a[:,2]>=90)),len(np.argwhere(a[:,3]>= 90))]
count=np.array(Count)

18. print count of students in each subject who got marks >= 90 print("Count of

students in each subject who got marks >= 90:", count)

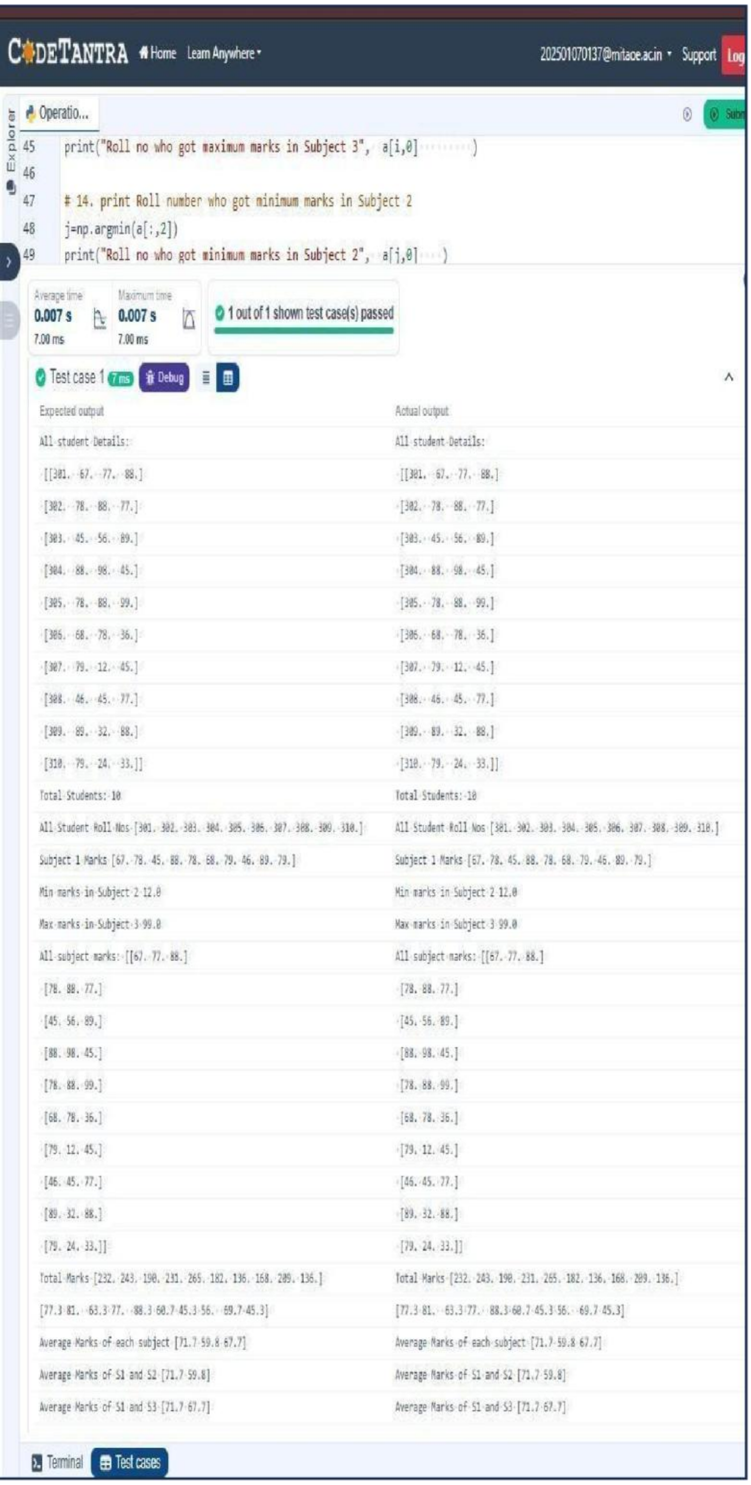
19. print count of subjects in which each student got marks >= 90 print("Roll no:", a[:,0]

) print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:]>=90,axis=1)

)

(a[:,1]<=90)]) k=np.where(a[:,1]==79)

print(k)



20. Print S1 marks in ascending order print(np.sort(a[:,1])) print(a[(a[:,1]>=50) & (a[:,1]<=90)]) print(a[(a[:,1]>=45) &